# Aryan Basnet

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## EDUCATION

**Rato Bangala School**, Patan Dhoka, Lalitpur **Jun 2022 – Jun 2024**
Cambridge International A Levels — Mixed Science
**Subjects:** Physics (9702), Chemistry (9701), Mathematics (9709), Computer Science (9618), English General Paper (8021)
**Final CAIE Grade:** A A A A b
**SAT:** 1550 / 1600

**Brihaspati Vidhyasadan School**, Naxal, Kathmandu **Apr 2016 – Apr 2022**
Secondary Education Examination (SEE)
**Subjects:** Science, Mathematics, Additional Mathematics, Computer Science, English, Nepali, Social Studies, Health, Population & Environment Education
**SEE GPA:** 3.70/4.00

## PROFESSIONAL EXPERIENCE

**National Policy Forum Nepal** **Jul 2026 – Present**
*Education Reform Research Intern*

- Conducted literature reviews and policy analysis on education reform, public education, and learning outcomes in Nepal.
- Supported the development of policy recommendations for the Ministry of Education focused on improving education delivery, classroom effectiveness, and student engagement.
- Designed and implemented a survey of high school teachers in Kathmandu Metropolitan City to assess teaching practices and institutional challenges.
- Assisted in preparing policy briefs, research reports, presentations, and analytical summaries related to education governance and reform.
- Published a policy blog examining the integration of Artificial Intelligence into Nepal's public education system.

**Hello World IT Solutions** **Dec 2022 - Mar 2023**
*Machine Learning Engineer*

- Developed a convolutional neural network (CNN) for detecting Pneumonia and Tuberculosis from 7,132 chest X-ray images.
- Achieved 94.5% diagnostic accuracy using a compact architecture with 3.7 million parameters.

• Evaluated model performance using 10-fold cross-validation, achieving 94.3% ± 1.0% across validation folds.
• Collaborated with medical experts to validate model predictions and assess the model's potential for medical screening.

**Billion Bricks** **Aug 2023 - Dec 2023**
*Product Design Intern*

• Collaborated directly with the CEO to design digital assets for the World Trade Organization Ministerial Conference.
• Conducted user research on housing needs to inform sustainable and affordable housing prototypes.
• Developed 3D housing concepts and prototypes using Blender and Sketch based on user research.

**The New York Academy of Sciences — The Junior Academy** **May 2023 - Aug 2023**
Student *Researcher*

• Researched algae blooms and lake acidification and their environmental and economic consequences.
• Analyzed the effects of large-scale algae blooms on tourism and local economies.
• Explored and proposed approaches for detecting, mitigating, and restoring affected lakes.

## LEADERSHIP EXPERIENCE

**Vidhata** **Jun 2023 – Present**
*Founder & Chief Executive Officer*

• Founded and lead a nonprofit organization providing structured non-academic and skill-based learning to foster holistic development among underserved students across Nepal.
• Built and coordinated a network of 125+ volunteers to deliver community-based learning programs.
• Reached 12,000+ students through structured educational sessions and workshops.
• Delivered 450+ hours of learning focused on practical and life skills.
• Established partnerships with 30 schools and NGOs to expand educational programs across communities.

**Japan-Nepal Health and Tuberculosis Research Association** **Oct 2022 - Apr 2023**
*Project Coordinator — Urban Tuberculosis Control Initiative*

• Coordinated an urban TB control initiative focused on active TB case finding using AI-assisted digital X-rays and TB-LAMP testing.
• Mobilized and coordinated Female Community Health Volunteers (FCHVs) for

community outreach, case identification, and patient follow-up.
- Organized local review meetings, trainings, and orientations to strengthen TB awareness and community-level coordination.
- Supported improvements to Urban DOTS clinic services, including coordination of essential equipment and patient-friendly care.

**Foundations of Machine Learning: A Comprehensive Guide for High Schoolers**
**Sep 2023 - Dec 2023**
*Co-author and Publisher*

- Co-authored a book introducing high school students to machine learning concepts, algorithm design, and practical applications.
- Designed the book cover and developed the final visual layout.
- Published the book on Amazon Kindle, reaching 175+ readers.

**Rato Bangala School Quiz Club** **Jul 2022 - Jun 2024**
*Founder & Captain*

- Founded and captained Rato Bangala School's first-ever General Knowledge Quiz Team.
- Conducted twice-weekly tutoring sessions to prepare team members for national competitions.
- Led the team to First Runner-Up among 298 schools nationwide at the Budhanilkantha School National Quiz Competition.
- Received a trophy, medal, and NPR 30,000 cash prize from former Prime Minister KP Sharma Oli.
- Secured coverage from four television media outlets following the team's achievement.

**Rato Bangala School Programming & Robotics Club** **Feb 2023 - Jun 2024**
*Vice President*

- Conducted twice-weekly programming sessions covering basic machine learning algorithms and model-building concepts through coding exercises.
- Collaborated with club members to engineer a sign-to-text glove capable of detecting hand gestures and converting them into written text.
- Represented the club at a regional Science Expo and presented the sign-to-text glove.
- Received the Best Project Award at the Science Expo.

## FELLOWSHIPS & PROFESSIONAL DEVELOPMENT

**Nabil SSE Certificate Course in Social Entrepreneurship** **Jun 2025 – Dec 2025**
*Social Entrepreneurship Fellow*

- Selected from 700+ applicants for a highly selective 25-member cohort focused on social entrepreneurship, innovation, and impact-driven ventures.

- Co-developed a research-informed and economically sustainable venture addressing rural-to-urban migration in Nepal.
- Participated in peer-learning sessions, policy simulations, and impact-scaling discussions with changemakers, social entrepreneurs, academics, and industry leaders from across Nepal.
- Received one-on-one mentorship from experienced social entrepreneurs and industry leaders to refine venture strategy, implementation, and impact goals.
- Developed practical skills in impact measurement, fundraising, sustainable business planning, and public-private partnerships through hands-on coursework and field-based learning.

## HONORS & ACHIEVEMENTS

• **Glocal's 20under20 Glocal Teen Hero Nepal (2024)** — Recognized among Nepal's top young changemakers for leadership and social impact.

• **Winner, X-Fest Hackathon (2023)** — Secured first place in a 24-hour national hackathon for developing an innovative technology solution.

• **First Runner-Up, Budhanilkantha School National Quiz Competition (2023)** — Placed second among 298 schools nationwide.

• **Silver Medal, National Chemistry Olympiad (2024)** — Earned national-level recognition for excellence in chemistry.

• **Bronze Medal, The Cathedral International Mathematics Competition (2024)** — Recognized for outstanding performance in an international mathematics competition.

• **Duke of Edinburgh's International Award, Bronze Level** — Completed the internationally recognized personal development program involving community service, skills development, and physical recreation.

• **Best Project Award (Winner), RBS Science Expo** — Recognized for developing and presenting a sign-to-text glove that converts hand gestures into written text.

## SKILLS & INTERESTS

**Programming & AI:** Python, Machine Learning, Deep Learning, Convolutional Neural Networks, Computer Vision, Data Analysis, AI Research

**Research & Policy:** Literature Review, Policy Analysis, Survey Design, Data Interpretation, Research Writing, Policy Writing, Report Writing, Educational Research

**Design & Product:** Figma, Blender, Sketch, Graphic Design, UI/UX Design, Product Prototyping, Visual Communication

**Leadership & Management:** Team Leadership, Project Management, Community Outreach, Workshop Facilitation, Curriculum Development, Stakeholder Engagement

**Communication:** Public Speaking, Presentation Design, Technical Writing, Professional Communication

**Tools:** Microsoft Office, Google Workspace, Git/GitHub